

User Manual for Myosense Application

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Contents

1	System Requirements and Initial Setup	4
2	Application Overview	5
2.1	Data Acquisition Module	5
2.2	Visualization and Processing Module	7
3	Basic Set Up	8
3.1	Installing the Myo SDK	8
3.2	Preparing the Repository	9
3.3	Verifying the Myo Armband Manager	9
3.4	Launching the Application	9
3.5	Initial Testing	9
4	Contact and Usage Policy	9

List of Figures

1	Myosense app	4
2	Myo Armband Manager Application	4
3	Data Acquisition Module	5
4	Identification Data	5
5	Start and Clear Buttons	6
6	General and EMG files	6
7	Data Visualization Module	7
8	Sensor 5 Data Visualization	7
9	View and Process Buttons	8
10	Prediction and Process files	8

1 System Requirements and Initial Setup

Before using the application, it is essential to add the 64-bit Myo SDK to the system path. This ensures that the Myo armband can properly communicate with the software. Once the SDK has been correctly configured, locate the compressed repository provided in the associated GitHub project. Inside this repository, double click the executable file named Myosense to launch the application.

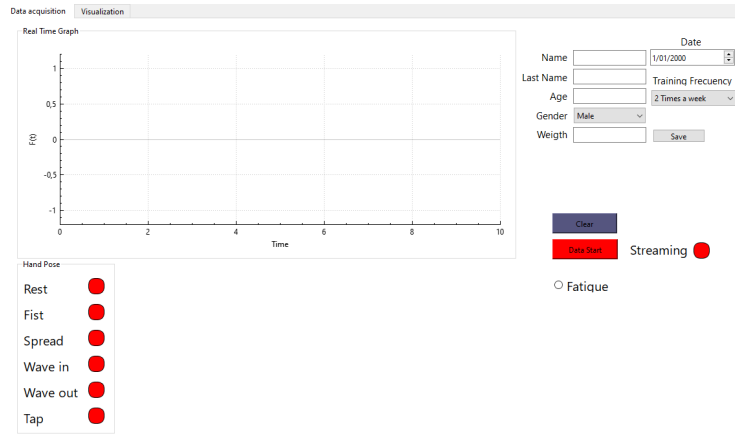


Figure 1: Myosense app

The system requires that the Myo armband is correctly placed on the subject and that the Myo Armband Manager application is running. This guarantees device recognition and data transmission.

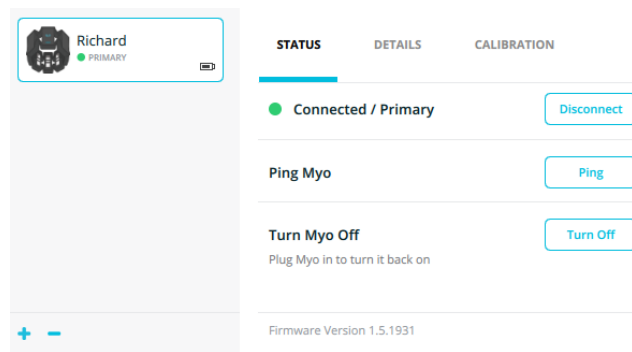


Figure 2: Myo Armband Manager Application

2 Application Overview

The application is structured into two main tabs: Data Acquisition and Visualization and Processing. These modules allow the user to collect raw EMG signals and subsequently analyze them through a variety of feature extraction methods.

2.1 Data Acquisition Module

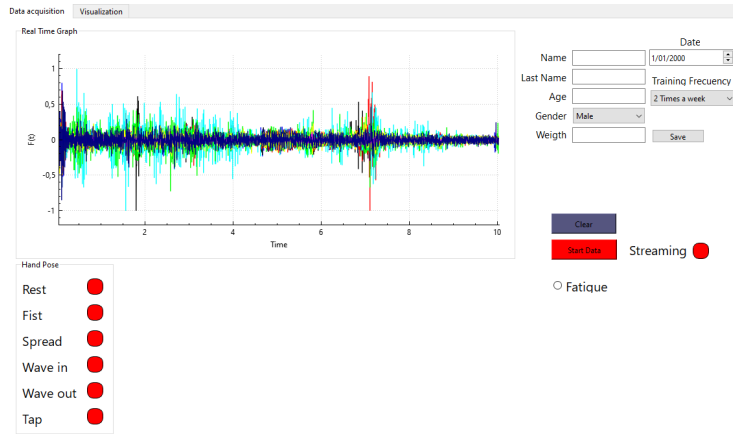


Figure 3: Data Acquisition Module

In the acquisition window, the user can create a folder dedicated to each participant. This folder stores both metadata and signal recordings. Metadata fields available within the interface include the subject's first name, last name, age, date of the session, weight used during the test, gender, and training frequency.

The screenshot shows the 'Identification Data' form. It includes fields for 'Name', 'Last Name', 'Age', 'Gender', and 'Weight'. The 'Date' field is set to '1/01/2000'. The 'Training Frequency' dropdown menu is set to '2 Times a week'. The 'Gender' dropdown menu is set to 'Male'. There is a 'Save' button at the bottom right.

Figure 4: Identification Data

After filling out these fields, pressing the Save button generates a folder named after the participant. Within this folder, a text file containing the entered metadata is created. Once the metadata has been stored, the Data Ac-

quisition button must be pressed.

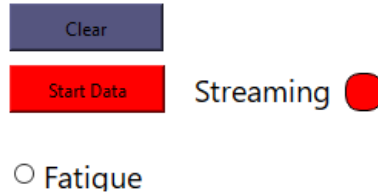


Figure 5: Start and Clear Buttons

The Myo armband provides haptic feedback through vibration, confirming its activation. From this moment, signals from the eight EMG sensors can be visualized in real time at a sampling frequency of 200 Hz.

To stop the acquisition process, the Data Acquisition button must be pressed again. Afterward, the Save button must be clicked once more to store the recorded signals. An important aspect of this process is the Fatigue toggle button. When the toggle is disabled, the saved signal file will be labeled No_Fatigue, and when the toggle is enabled, the corresponding file will be labeled Fatigue.

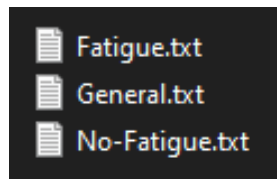


Figure 6: General and EMG files

2.2 Visualization and Processing Module

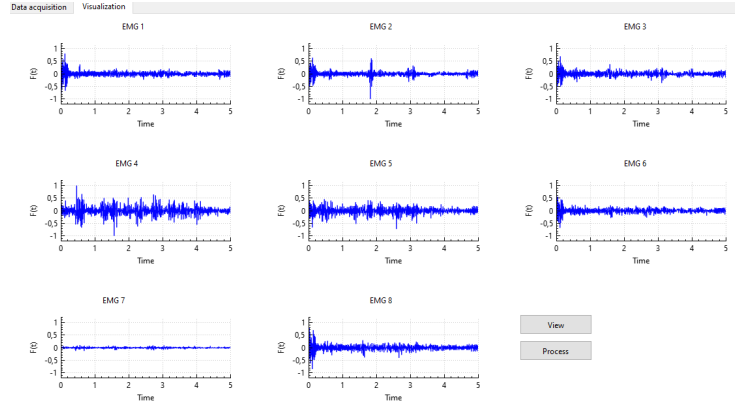


Figure 7: Data Visualization Module

The second tab, named Visualization, is dedicated to reviewing and processing the collected signals. This section displays eight empty plots by default, each corresponding to one of the Myo armband sensors.

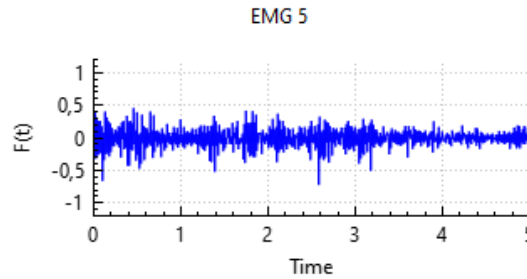


Figure 8: Sensor 5 Data Visualization

The View button updates all plots simultaneously, presenting the most recently stored signals. The Process button initiates the computational analysis of the signals.

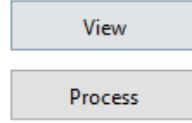


Figure 9: View and Process Buttons

During this step, the application generates multiple text files containing calculated features, which include entropy, entropy derivative, variance, mean absolute value, root mean square, and a dedicated file containing the fatigue prediction. The fatigue prediction file highlights the specific moments and the sensors that exhibited muscular fatigue during the test.

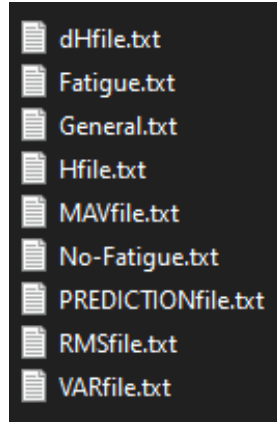


Figure 10: Prediction and Process files

All output files are saved in TXT format. Each file is structured in eight columns, with each column corresponding to one of the eight EMG channels of the Myo armband.

3 Basic Set Up

3.1 Installing the Myo SDK

The Myosense application requires the official Myo SDK (64-bit) for correct device communication. Download the SDK from the official Thalmic Labs repository and extract its contents into a local directory. Once extracted, locate the folder containing the SDK libraries and executables.

To enable system-wide recognition of the SDK, add the path of the extracted SDK to the environment variables of the operating system. On Windows, this is achieved by navigating to System Properties, selecting Advanced System Settings, opening Environment Variables, and modifying the Path variable. Insert the full directory path to the SDK's bin folder and confirm the changes.

3.2 Preparing the Repository

Access the GitHub repository associated with this project and download the compressed package containing the application files. Once downloaded, extract the contents into a preferred working directory. Inside the extracted folder, identify the Myosense executable file. This file serves as the main entry point to the application.

3.3 Verifying the Myo Armband Manager

Before launching Myosense, ensure that the Myo Armband Manager application is installed and running. This software manages the pairing and active state of the Myo device. Place the armband on the subject's forearm according to the manufacturer's guidelines, ensuring proper alignment with the muscle group under study. Verify that the device connects successfully and is visible in the Myo Armband Manager interface.

3.4 Launching the Application

To initiate the program, double click the Myosense executable. The application window will open, providing access to the two main tabs: Data Acquisition and Visualization and Processing. If the SDK and Myo Manager are correctly configured, the software will automatically detect the armband.

3.5 Initial Testing

Once inside the application, perform a preliminary test to validate correct configuration. Navigate to the Data Acquisition tab, enter test metadata, and press the Data Acquisition button. The armband should emit a vibration as haptic feedback. If the signals from the eight EMG sensors appear in real time at 200 Hz, the setup has been correctly executed. Stop the test by pressing the acquisition button again and confirm that the generated files are properly saved.

4 Contact and Usage Policy

For any questions, technical issues, or further clarifications regarding the Myosense application, you may contact the developer directly at est.juan.ramirez3@unimilitar.edu.co. Suggestions and recommendations are always welcome and will be carefully considered for future improvements.

The application is released as free-to-use software and is fully open to modifications and extensions by the research community or individual users. However, we kindly request that proper acknowledgment be given through citation if the application or its results are used in academic, professional, or research contexts. This ensures that the contribution is properly recognized while supporting collaborative development and continuous enhancement of the tool.